ChurnWatch AI

AI Product Development and Monitoring

PROJECT REPORT

A churn-risk assessment MVP combining rule-based risk scoring, LLM-assisted analysis, and monitoring/evaluation.

1. Executive Summary

ChurnWatch AI is an AI-assisted churn-risk analysis MVP. It combines a transparent, rule-based risk assessment with an LLM-generated narrative and retention recommendations. A separate Monitoring API is intended to collect request metadata and persist it in Supabase PostgreSQL. The Monitoring UI presents operational performance and historical evaluation metrics.

The supplied assignment requires three components: a working AI product, a development environment, and monitoring/evaluation connected to product behavior. It also requires two technical diagrams, tool decisions, benchmarks/evaluation results, references, a source-code handoff, and a pitch deck.

Component	Purpose	Evidence to verify
AI Product	Customer risk assessment and AI-assisted insights/recommendations.	Working analysis endpoint and UI demo.
Development Environment	Repository, local setup, API docs, tests and Git workflow.	Screenshots, commands and actual test output.
Monitoring & Evaluation	Health, latency, tokens, errors and historical risk evaluation.	Live event persistence and valid metric calculations.

Core integrity rule

Successful LLM execution is not proof of correct churn-risk assessment. Risk scores are heuristic points, not calibrated probabilities. Evaluation must use correctly matched observable outcomes and disclose sample sizes.

2. Problem Statement and Objectives

Problem: Customer churn creates a need to identify customers who may be at elevated risk. A risk score can prioritize review, while an LLM can convert customer context into readable insights and retention actions. Monitoring is needed to distinguish technical availability from useful evaluation performance.

Objective: Retrieve customer data, calculate the existing risk category, generate AI-assisted analysis, record operational events, and compare risk categories with observed historical churn where the dataset supports valid comparisons.

- Allow a user to request customer analysis and receive insights/recommendations or a clearly labelled fallback if the LLM fails.
- Track real request counts, latency, token usage and failures when available.
- Evaluate 30-day and 90-day outcomes separately, exclude unobserved outcomes, and show denominators/sample sizes.
- Avoid claiming heuristic points are probabilities or that historical results guarantee future performance.

3. Product and Technical Architecture

Production frontend → Main API → customer dataset and risk rules → Groq LLM → analysis response. The Main API submits a monitoring event to the Monitoring API. The Monitoring API stores request metadata in Supabase PostgreSQL. The Monitoring frontend reads monitoring summaries/history and displays evaluation results.

Component	Responsibility	Boundary / caution
Production Frontend	Dashboard, Customer Explorer, Customer Details, AI Analysis.	Calls APIs; no secrets or direct database access.
Main API	Customer data, risk rules, analysis endpoint and Groq integration.	Existing risk logic remains the source of truth.
Churn dataset	Customer features/snapshots and historical outcomes.	Verify actual file names, columns and outcome semantics.
Groq LLM	Generates narrative insights and retention recommendations.	Availability is not correctness.
Monitoring API	Ingests events and serves summary/request history.	Monitoring failure should not unnecessarily break analysis.
Supabase PostgreSQL	Persists monitoring events and metadata.	Verify live schema, RLS, connection and write/readback.
Monitoring Frontend	Shows operational metrics and historical evaluation.	Use real API data and explicit unavailable states.

Risk-score interpretation

Project notes describe heuristic thresholds Low = 0–2, Medium = 3–5, High = 6+. Verify these against the actual implementation. Do not describe the score as a churn probability unless a calibrated probability model is implemented and validated.

4. Data Collection and Preparation

The project context identifies a synthetic customer churn CSV, previously supplied as WA_Fn-UseC_-Telco-Customer-Churn.csv. Earlier inspection notes reported 5,000 customers, 70,575 snapshots and 5,000 outcomes; recheck these counts against the current files before final submission. Do not describe the dataset as real bank data unless its source proves that.

- Record exact source, filename, licensing/source URL, row count, customer count, snapshot count and target definitions.
- Inspect data types, missing values, duplicate keys, date ranges and class balance.
- Identify the exact risk inputs, score/category, 30-day outcome and 90-day outcome columns; do not guess names.
- Match each risk assessment to the correct customer snapshot and subsequent outcome; avoid future-data leakage.
- Treat null, missing, unobserved or immature outcomes as unevaluable—not as retained customers.
- Document preprocessing and exclusions so the evaluation can be reproduced.

Data provenance limitation: The assignment asks for data collection to be documented and external datasets cited. Insert the verified dataset source URL and license after checking the actual file origin. Counts above are prior project notes, not newly measured results.

5. Monitoring Metrics and Evaluation

The supplied instructions name four monitoring types: Application Monitoring, Performance Monitoring, AI Quality Monitoring, and Cost Monitoring. ChurnWatch AI maps these to request health, latency, historical risk evaluation, and token consumption.

Type	Metric	Definition / value	Source
Application	Request count / error rate	Shows volume and proportion of failed requests.	Monitoring records
Performance	Latency	Shows response time and potential slowdowns/timeouts.	Per-request metadata
AI quality	High-Risk Precision	High-Risk records with observed churn ÷ High-Risk records with observable outcomes.	Risk + outcome
AI quality	High-Risk Recall	High-Risk observed churn cases ÷ all observed churn cases across categories.	Risk + outcome
AI quality	Churn rate by risk	Observed churn ÷ eligible records in each category; 30/90 days separate.	Historical dataset
AI quality	Evaluation coverage	Evaluated eligible records ÷ total eligible records, with rules documented.	Evaluation pipeline
Cost	Token usage	Prompt, completion and total tokens; monetary cost requires verified pricing.	LLM metadata
Reliability	LLM success/failure	Separates provider reliability from prediction quality.	LLM metadata

Exact formulas

High-Risk Precision = High-Risk records with observed churn / High-Risk records with observable outcomes × 100.

High-Risk Recall = High-Risk records with observed churn / all eligible records with observed churn × 100.

Churn rate by category = observed churn cases in category / records in category with observable outcomes × 100.

Evaluation coverage = unique eligible assessment records with observable outcomes / unique records eligible for the horizon × 100.

Always show numerator, denominator, horizon and sample size. If no eligible positive cases exist, display N/A rather than 0%. Define the unit of analysis because multiple snapshots may exist per customer.

Evaluation limitations

Use 30-day and 90-day outcomes separately. Retrospective results show alignment with historical outcomes, not guaranteed prospective performance, causal impact, or correctness of LLM narratives. If outcome matching is unreliable, show Insufficient Data and explain the blocker.

6. Benchmark and Differentiation

1. Feature Benchmark

✓ Documented capability | ◐ Depends on configuration / feature | Not established here ≠ unavailable

Feature	ChurnZero	Salesforce Einstein	Adobe Customer AI	ChurnWatch AI
Churn / customer risk identification	✓	✓	✓	✓
Risk signals or contributing factors	✓	◐	✓	✓
Customer-facing explanation / insights	✓	◐	✓	✓ Groq-generated
Transparent, rule-based points	Not established	Not established	Not established	✓
Retention recommendations	✓ / feature-dependent	◐	Not established	✓ Groq-generated
LLM tokens, latency and errors monitoring	Not established	Not established	Not established	✓ intended; verify live data
Historical Precision / Recall in this project	Requires verification	Requires verification	Requires verification	Planned / measure on labelled outcomes

2. What ChurnWatch AI Adds

Addition	Project value
Transparent rule-based scoring	Shows the risk factors behind a category; points are not a churn probability.
Separate scoring from LLM explanation	Backend calculates the score; Groq explains the supplied evidence and suggests actions.
LLM operational monitoring	Tracks request status, latency, token usage and errors when actual events are recorded.
Outcome-based evaluation	Measures High-Risk Precision, High-Risk Recall and observed churn by risk category when outcomes are available.

3. Evaluation Metrics

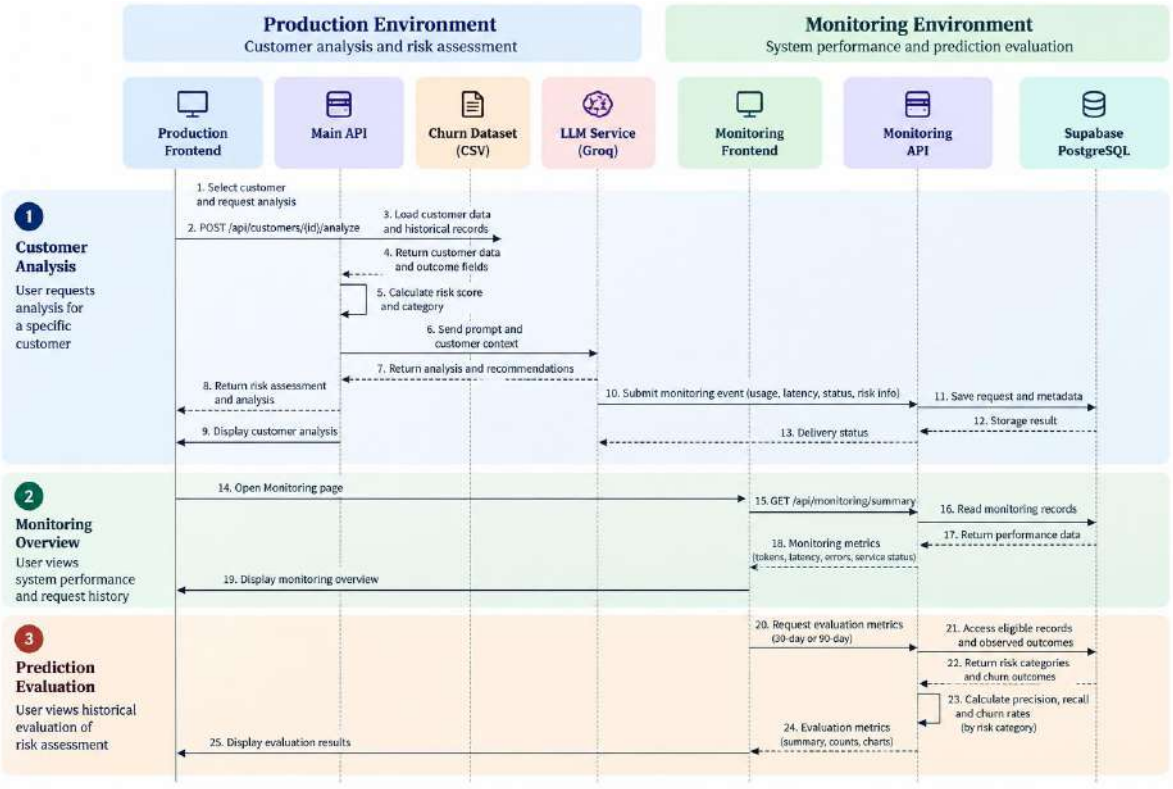
Metric	Calculation
High-Risk Precision	High-risk records that churned ÷ high-risk records with observed outcomes.
High-Risk Recall	High-risk records that churned ÷ all eligible observed churn records.
Churn rate by risk level	Observed churn in Low / Medium / High ÷ records with observed outcomes in that level.

Evaluate 30-day and 90-day outcomes separately when available. Show sample counts; missing or unobserved outcomes must not be counted as retained.

7. Technical Diagrams

The assignment explicitly requires two diagrams: Development Flow and Monitoring Flow. The sequence diagram below is the Monitoring Flow. Check it against the final implementation and endpoints before submitting.

7.1 Monitoring Flow — Sequence Diagram



Note: Supabase event storage does not automatically mean historical outcomes are available for evaluation. Verify the evaluation data path.

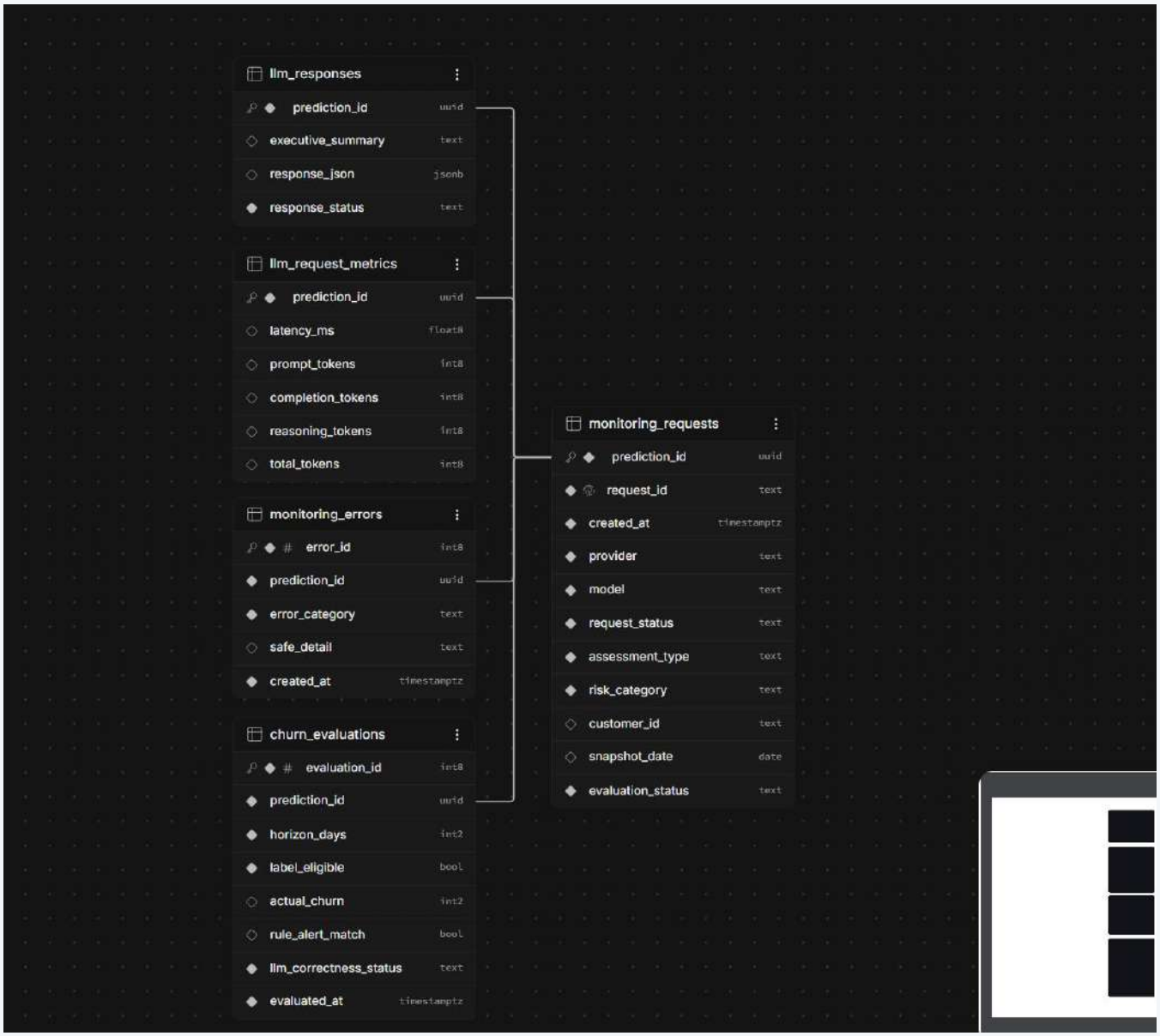
7.2 Development Flow

Problem definition → dataset inspection and target definition → risk rule/Main API implementation → Groq integration and fallback → monitoring event schema/persistence → frontend and monitoring UI → API/data/UI tests → verification, documentation and Git review.

Development stage	Evidence to capture
Define problem and success criteria	Document problem, intended user and success criteria.
Inspect dataset and define horizons	Record source, schema, target definitions and eligibility rules.
Implement/maintain risk rules and Main API	Code review and API tests.
Integrate Groq and fallback	Success/failure tests and truthful fallback labels.
Monitoring ingestion and persistence	Event schema, database insert/readback evidence.
Frontend and Monitoring UI	Screenshots of each real tab and API-backed values.
Test and verify	Actual test/build output, metric calculations, edge cases.
Review and document	Git diff, known limitations, results and references.

8. Supabase Schema and Database Verification

Attach the actual Supabase schema screenshot below. Document real table names, columns, keys, indexes and RLS policies from the applied schema—not only from migration files.



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Database verification checklist

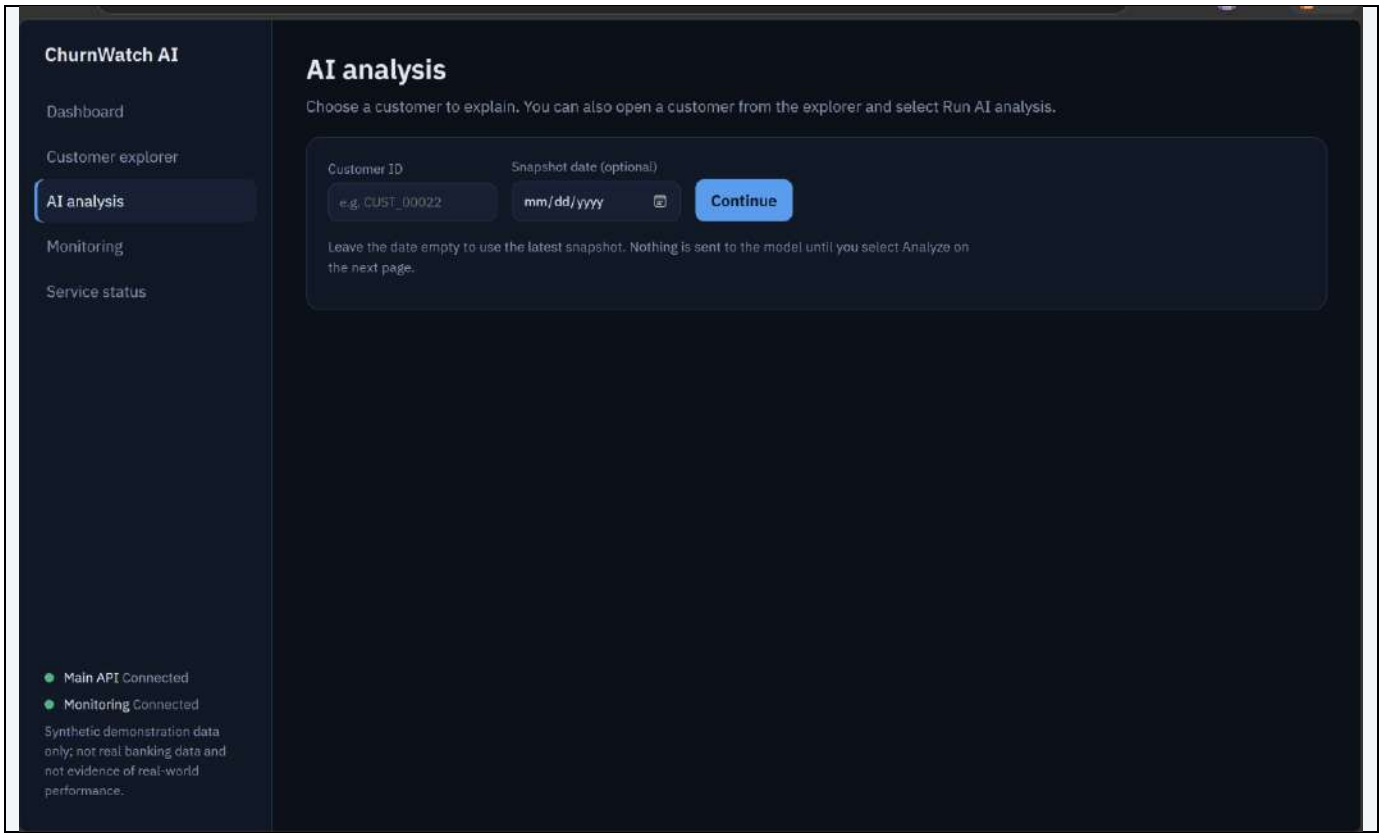
- Confirm migrations were applied in the correct order.
- Confirm Monitoring API can connect to PostgreSQL with the required SSL/connection method.

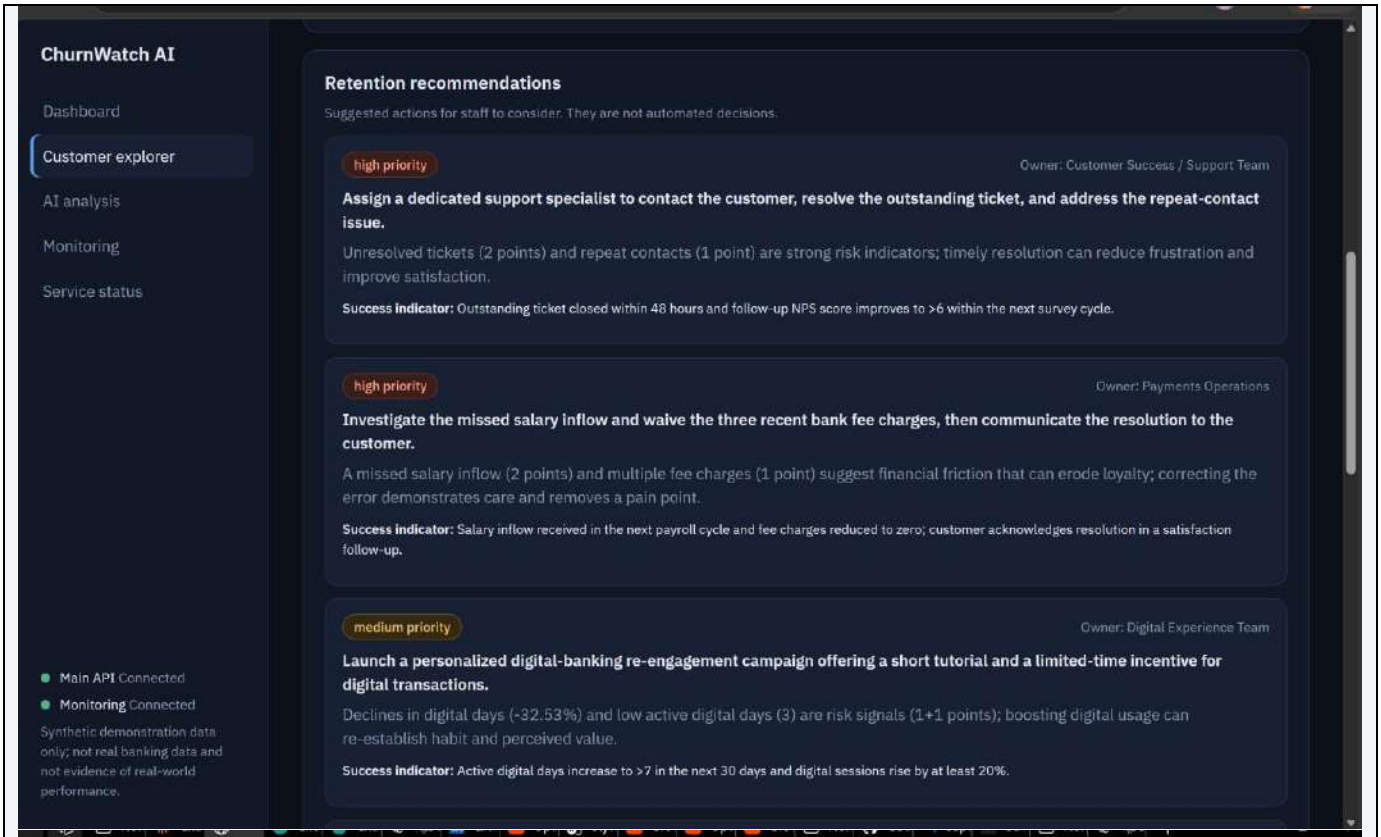
- Insert a test event and verify it can be read back through the Monitoring API.
- Verify RLS permits intended backend operations without disabling security.
- Never include DATABASE_URL, passwords, API keys or service-role secrets in screenshots.

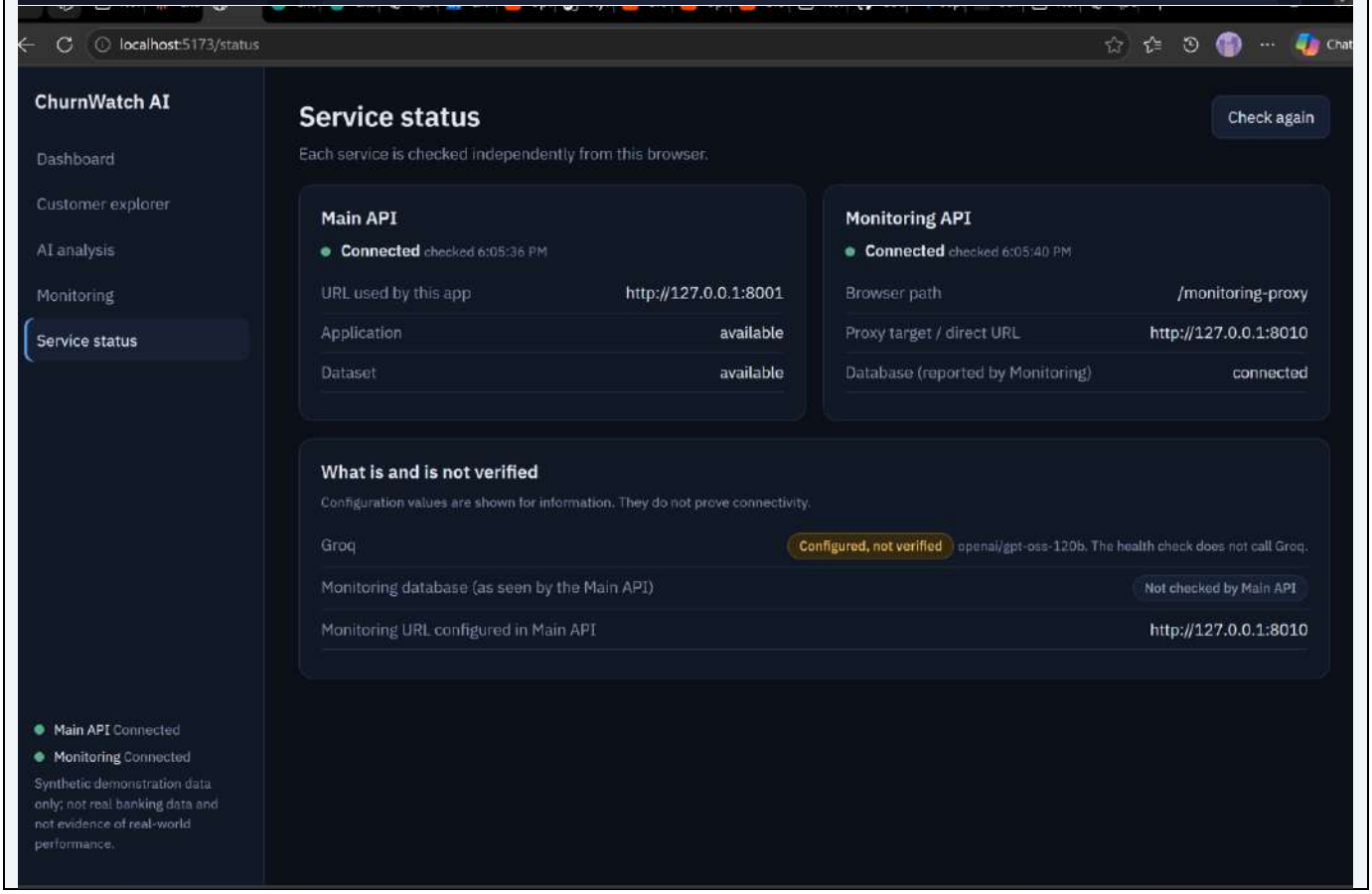
Verification status: available materials do not prove live schema, connectivity, RLS or persistence has passed. Mark each check Passed/Failed/Pending only after testing.

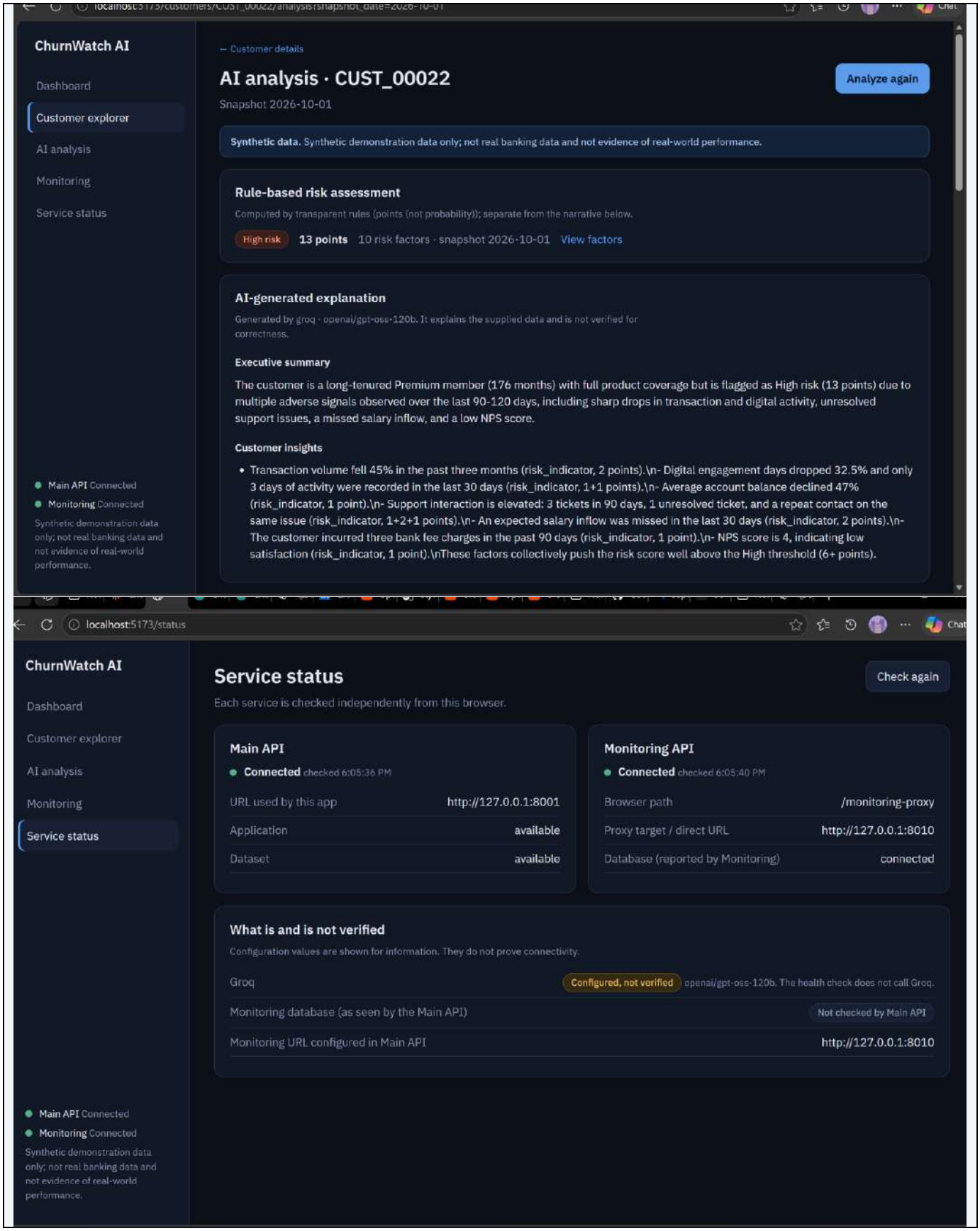
9. System Screenshots

Attach real screenshots from the running application. Redact sensitive identifiers and secrets.

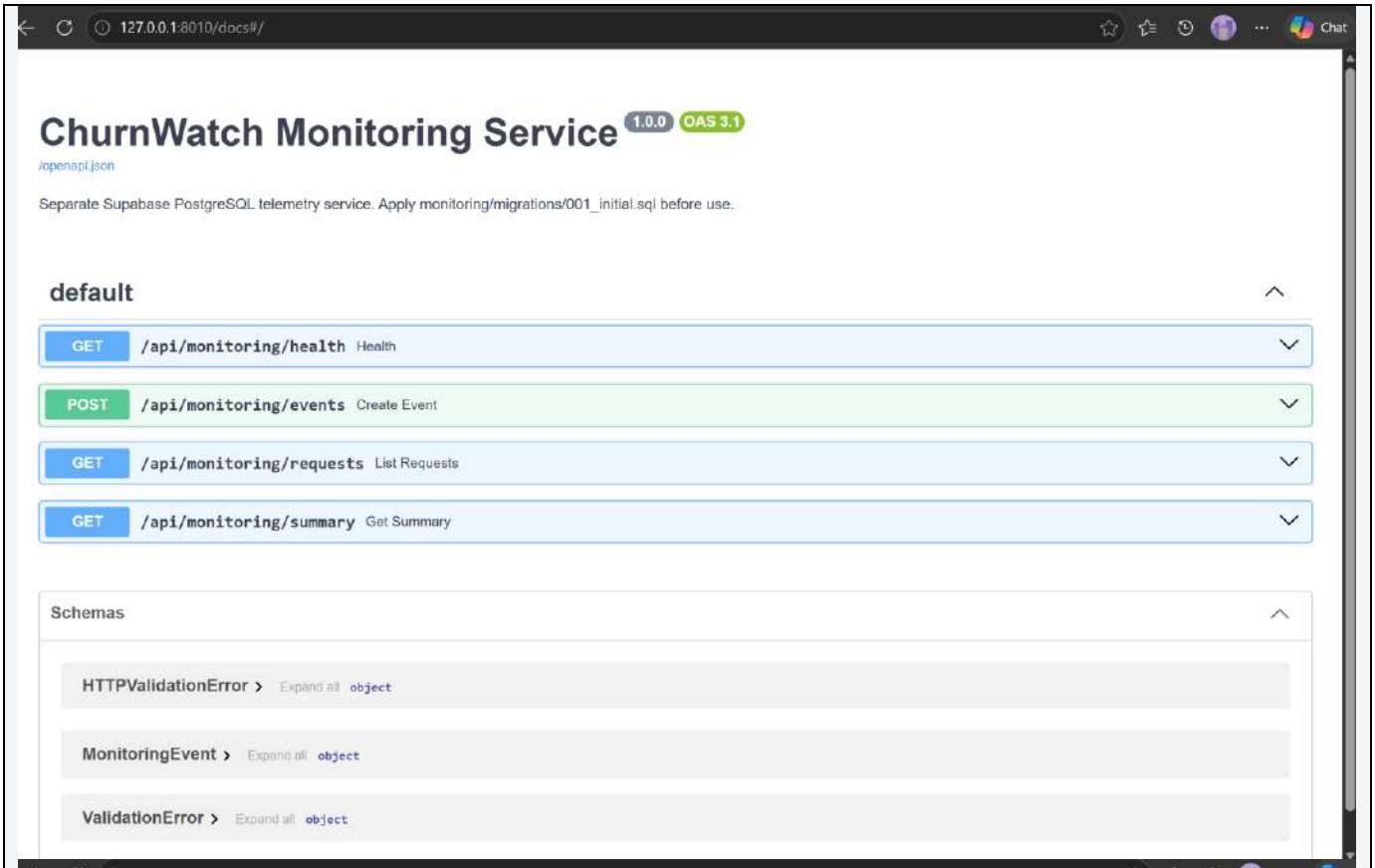


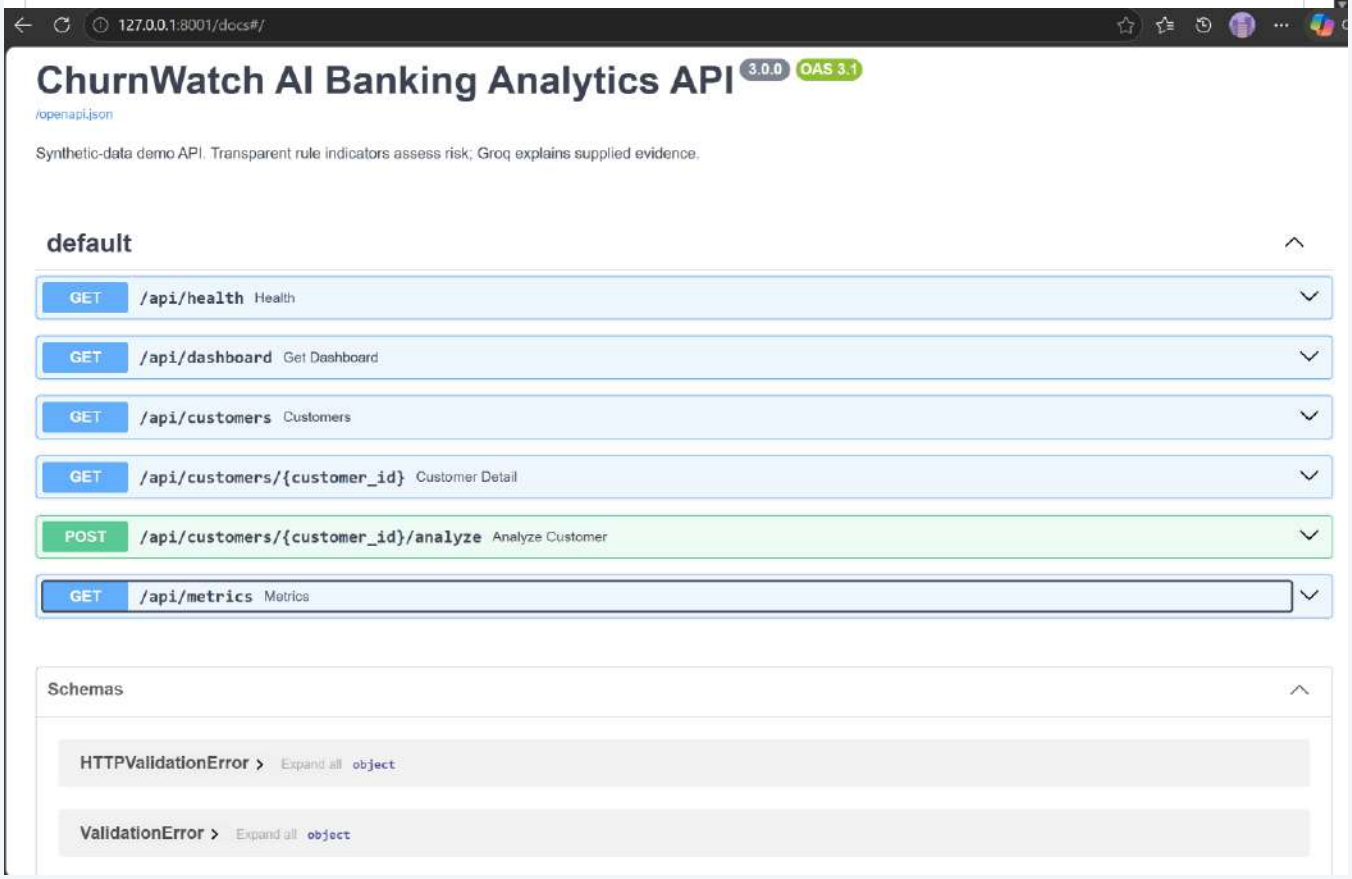

ChurnWatch AI
Dashboard
Customer explorer
AI analysis
Monitoring
Service status
Main API Connected
Monitoring Connected
Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
Retention recommendations
Suggested actions for staff to consider. They are not automated decisions.
high priority
Owner: Customer Success / Support Team
Assign a dedicated support specialist to contact the customer, resolve the outstanding ticket, and address the repeat-contact issue.
Unresolved tickets (2 points) and repeat contacts (1 point) are strong risk indicators; timely resolution can reduce frustration and improve satisfaction.
Success indicator: Outstanding ticket closed within 48 hours and follow-up NPS score improves to >6 within the next survey cycle.
high priority
Owner: Payments Operations
Investigate the missed salary inflow and waive the three recent bank fee charges, then communicate the resolution to the customer.
A missed salary inflow (2 points) and multiple fee charges (1 point) suggest financial friction that can erode loyalty; correcting the error demonstrates care and removes a pain point.
Success indicator: Salary inflow received in the next payroll cycle and fee charges reduced to zero; customer acknowledges resolution in a satisfaction follow-up.
medium priority
Owner: Digital Experience Team
Launch a personalized digital-banking re-engagement campaign offering a short tutorial and a limited-time incentive for digital transactions.
Declines in digital days (-32.53%) and low active digital days (3) are risk signals (1+1 points); boosting digital usage can re-establish habit and perceived value.
Success indicator: Active digital days increase to >7 in the next 30 days and digital sessions rise by at least 20%.


localhost:5173/status
ChurnWatch AI
Dashboard
Customer explorer
AI analysis
Monitoring
Service status
Main API Connected
Monitoring Connected
Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
Service status
Each service is checked independently from this browser.
Check again
Main API
Connected checked 6:05:36 PM
URL used by this app http://127.0.0.1:8001
Application available
Dataset available
Monitoring API
Connected checked 6:05:40 PM
Browser path /monitoring-proxy
Proxy target / direct URL http://127.0.0.1:8010
Database (reported by Monitoring) connected
What is and is not verified
Configuration values are shown for information. They do not prove connectivity.
Groq Configured, not verified openai/gpt-oss-120b. The health check does not call Groq.
Monitoring database (as seen by the Main API) Not checked by Main API
Monitoring URL configured in Main API http://127.0.0.1:8010

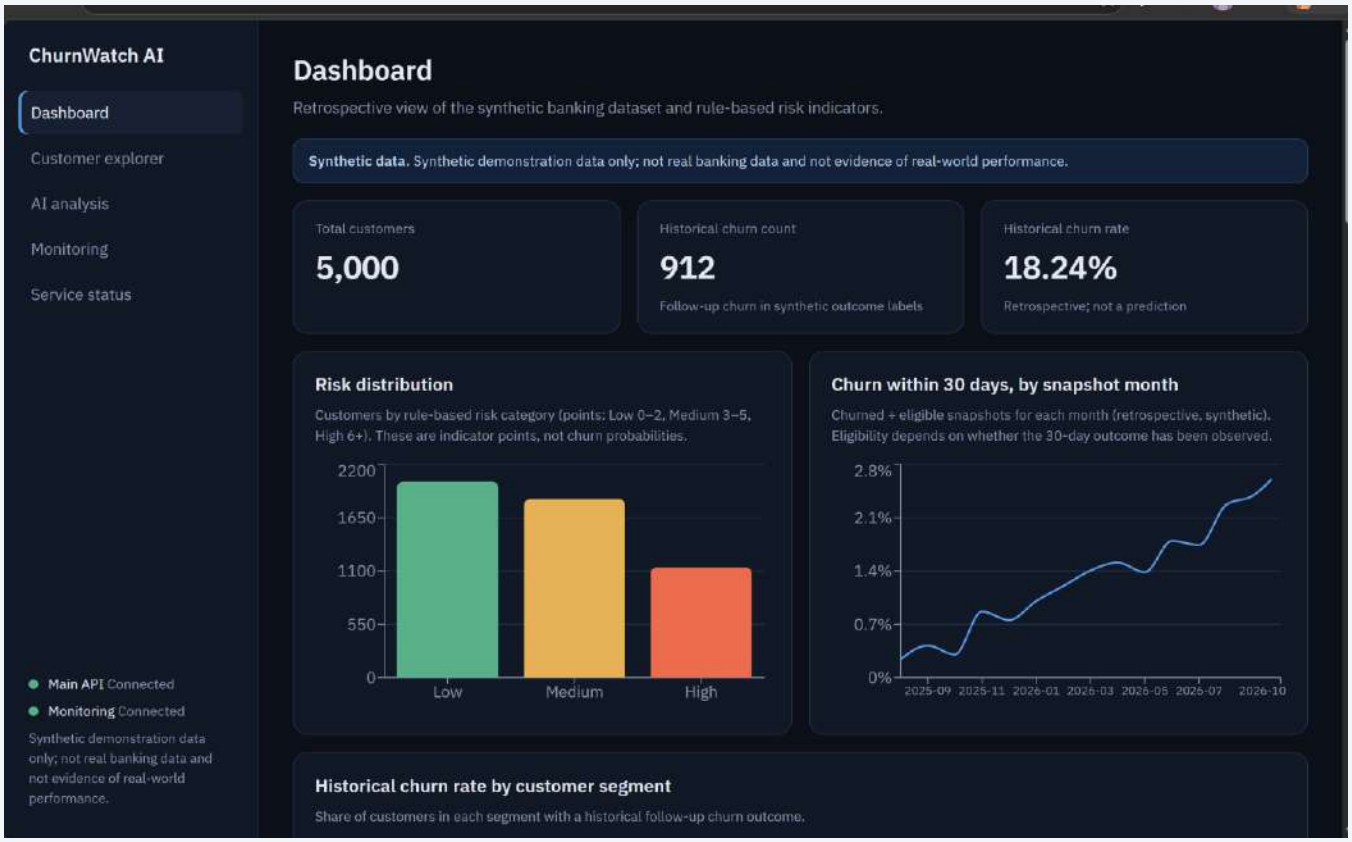

ChurnWatch AI
Dashboard
Customer explorer
AI analysis
Monitoring
Service status
Main API Connected
Monitoring Connected
Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
← Customer details
AI analysis · CUST_00022
Snapshot 2026-10-01
Analyze again
Synthetic data. Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
Rule-based risk assessment
Computed by transparent rules (points (not probability)); separate from the narrative below.
High risk
13 points
10 risk factors · snapshot 2026-10-01
View factors
AI-generated explanation
Generated by groq · openai/gpt-oss-120b. It explains the supplied data and is not verified for correctness.
Executive summary
The customer is a long-tenured Premium member (176 months) with full product coverage but is flagged as High risk (13 points) due to multiple adverse signals observed over the last 90-120 days, including sharp drops in transaction and digital activity, unresolved support issues, a missed salary inflow, and a low NPS score.
Customer insights
• Transaction volume fell 45% in the past three months (risk_indicator, 2 points).\n- Digital engagement days dropped 32.5% and only 3 days of activity were recorded in the last 30 days (risk_indicator, 1+1 points).\n- Average account balance declined 47% (risk_indicator, 1 point).\n- Support interaction is elevated: 3 tickets in 90 days, 1 unresolved ticket, and a repeat contact on the same issue (risk_indicator, 1+2+1 points).\n- An expected salary inflow was missed in the last 30 days (risk_indicator, 2 points).\n- The customer incurred three bank fee charges in the past 90 days (risk_indicator, 1 point).\n- NPS score is 4, indicating low satisfaction (risk_indicator, 1 point).\nThese factors collectively push the risk score well above the High threshold (6+ points).
localhost:5173/status
ChurnWatch AI
Dashboard
Customer explorer
AI analysis
Monitoring
Service status
Main API Connected
Monitoring Connected
Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
Service status
Each service is checked independently from this browser.
Check again
Main API
Connected checked 6:05:36 PM
URL used by this app
http://127.0.0.1:8001
Application
available
Dataset
available
Monitoring API
Connected checked 6:05:40 PM
Browser path
/monitoring-proxy
Proxy target / direct URL
http://127.0.0.1:8010
Database (reported by Monitoring)
connected
What is and is not verified
Configuration values are shown for information. They do not prove connectivity.
Groq
Configured, not verified
openai/gpt-oss-120b. The health check does not call Groq.
Monitoring database (as seen by the Main API)
Not checked by Main API
Monitoring URL configured in Main API
http://127.0.0.1:8010

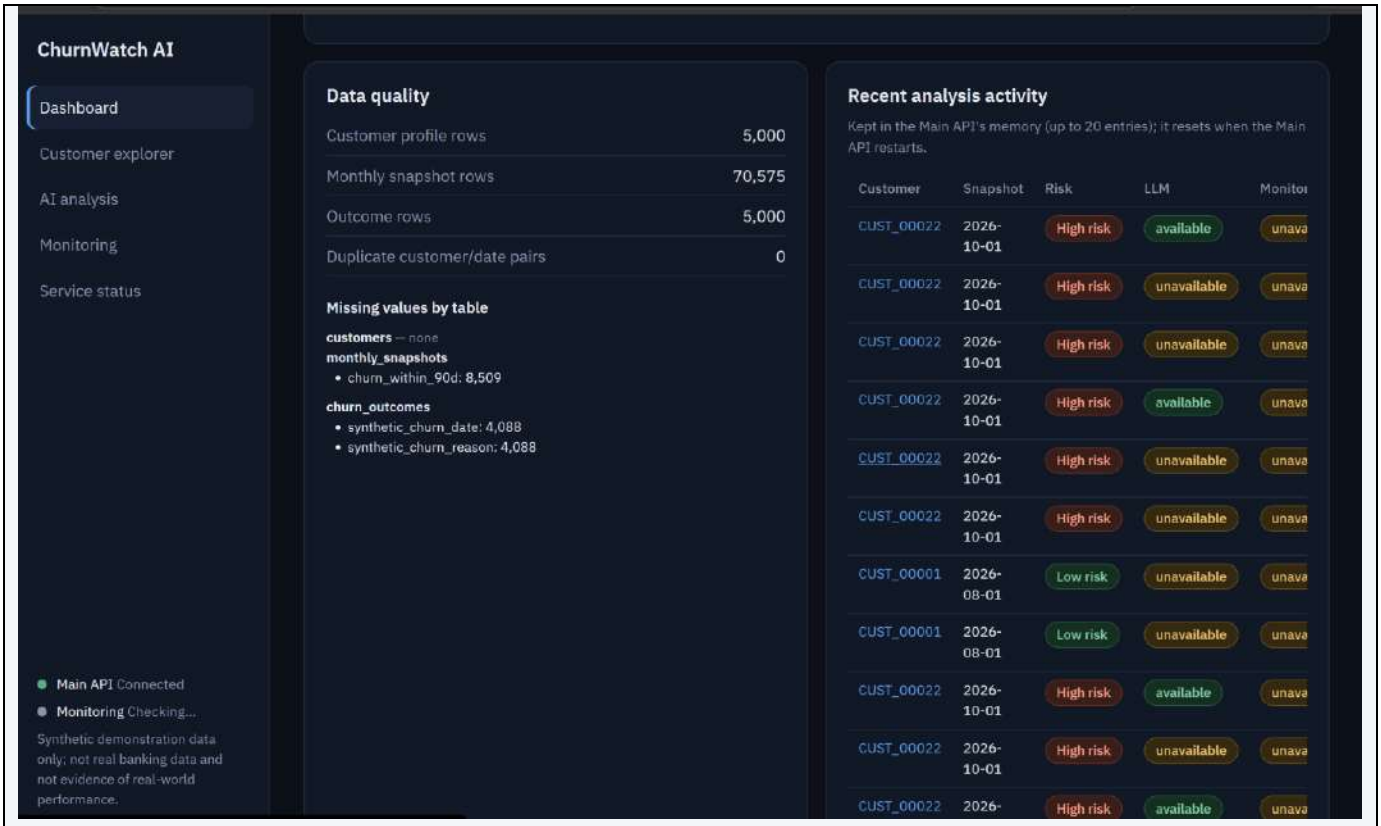

localhost:5173/monitoring
ChurnWatch AI
Dashboard
Customer explorer
AI analysis
Monitoring
Service status
Main API Connected
Monitoring Connected
Synthetic demonstration data only; not real banking data and not evidence of real-world performance.
Monitoring
Refresh
LLM request history and summary from the separate Monitoring service.
Outcome horizon
30 days
Weak historical performance — 30d horizon. High-Risk precision is 3.4% (198 of 5,817); recall is 21.6% (198 of 915 observed churn records).
Provisional interpretation: strong requires precision and recall ≥60%; weak means both are <40%; otherwise performance is mixed. Conclusions require at least 100 evaluated records, 30 High-Risk records, and 30 observed churn outcomes. Risk categories use the existing heuristic thresholds: Low 0–2, Medium 3–5, High 6+ points.
High-Risk Precision
3.4%
198 churned of 5,817 High-Risk records
High-Risk Recall
21.6%
198 High-Risk among 915 observed churn records
Historical churn rate by risk category
Observed 30d snapshot outcomes only. Bars show churned / evaluated records; risk points are not probabilities.
3.6%
2.7%
1.8%
0.9%
Evaluation coverage
Coverage is the share of unique customer/snapshot records with an observed outcome at this horizon.
70,575 evaluated of 70,575 records (100%).
Missing or unavailable outcome 0
Unobserved horizon 0
Missing label 0
Unique customers represented 5,000


127.0.0.1:8010/docs#/
Chat
ChurnWatch Monitoring Service
1.0.0
OAS 3.1
/openapi.json
Separate Supabase PostgreSQL telemetry service. Apply monitoring/migrations/001_initial.sql before use.
default
GET /api/monitoring/health Health
POST /api/monitoring/events Create Event
GET /api/monitoring/requests List Requests
GET /api/monitoring/summary Get Summary
Schemas
HTTPValidationError > Expand all object
MonitoringEvent > Expand all object
ValidationError > Expand all object


127.0.0.1:8001/docs#/
ChurnWatch AI Banking Analytics API
3.0.0
OAS 3.1
/openapi.json
Synthetic-data demo API. Transparent rule indicators assess risk; Groq explains supplied evidence.
default
GET /api/health Health
GET /api/dashboard Get Dashboard
GET /api/customers Customers
GET /api/customers/{customer_id} Customer Detail
POST /api/customers/{customer_id}/analyze Analyze Customer
GET /api/metrics Metrics
Schemas
HTTPValidationError > Expand all object
ValidationError > Expand all object

. System Screenshots (continued)





10. AI Tools and Technical Decisions

Tool	Role	Rationale / trade-off
Python + FastAPI	Main and Monitoring APIs	Lightweight services with clear separation of product and monitoring duties.
Groq API / configured LLM	Generates insights and retention recommendations	Avoids training/serving a custom model for MVP; output still needs evaluation.
TypeScript frontend	Typed web UI for dashboard, customers, analysis and monitoring	Type safety helps maintain API contracts; confirm framework/versions from package files.
Flutter (reported by owner)	Include only if a Flutter client/module exists in submitted implementation	Verify its role in repository; do not claim a Flutter app exists if only TypeScript web UI is present.
Supabase PostgreSQL	Stores monitoring events and metadata	Managed PostgreSQL; verify schema, connection, RLS and readback.
VS Code / Git / Codex / Claude	Development and AI-assisted coding as used	Document which work was assisted and manually reviewed; verify with tests and code inspection.

AI literacy and verification

AI coding tools may assist with code generation, troubleshooting and UI scaffolding. The team remains responsible for understanding, reviewing and testing generated code, checking API contracts, verifying metric formulas and documenting limitations. Identify AI-assisted versus manual work based on the actual development history.

11. Limitations and Mitigations

- Synthetic data may not generalize to real banking customers; label the demo accordingly.

- Retrospective evaluation is not proof of prospective effectiveness.
- Heuristic points are not calibrated probabilities; do not claim probability calibration.
- 30-day and 90-day labels require adequate follow-up; unobserved outcomes are not negative outcomes.
- LLM variability means provider success does not prove correctness.
- Monitoring outages should be visible without unnecessarily breaking customer analysis.
- Protect database credentials and retain RLS.
- No external numerical benchmark is claimed until a comparable source and method are documented.

12. Results to Populate After Evaluation

Do not fill this table with illustrative values. Populate it from a reproducible run on the current dataset and code.

Metric	30-day	90-day	Required note
High-Risk Precision	Pending	Pending	TP / (TP + FP), eligible records.
High-Risk Recall	Pending	Pending	TP / (TP + FN), observed churn cases.
Low/Medium/High churn rates	Pending	Pending	Counts and eligible denominator for each category.
Evaluation coverage	Pending	Pending	Define eligible unit and exclusions.
LLM success rate	Pending	N/A	Successful requests / eligible requests.
Mean latency	Pending	N/A	Unit, sample size and included requests.
Token usage	Pending	N/A	Prompt, completion, total; cost needs verified pricing.

Strong / Mixed / Weak / Insufficient Data labels should use documented provisional thresholds and minimum sample-size rules. These are internal project criteria, not industry standards.

13. References and Submission Checklist

Primary source: supplied “Project 1 instructions.pdf,” titled “AI Product Development and Monitoring.” It requires an AI product MVP, development environment, monitoring dashboard connected to product/supporting services, source-code handoff, a concise report documenting problem, solution, data collection, benchmarks, technical design, metrics, evaluation results, tools, diagrams and references, plus a pitch deck. The brief specifies a 3-minute pitch and 10-minute discussion/Q&A for Thursday 22 October 2026.

References to complete with exact URLs and access dates

- Dataset original source page and license for the exact CSV used.
- Supabase official documentation for PostgreSQL connections, pooling and RLS actually used.
- Groq API documentation for response fields and token usage actually used.
- Any external evaluation framework or benchmark actually used, with its role explained.

These are reference targets, not claims that the pages have already been consulted. Add exact URLs after verifying the sources used.

Final submission checklist

- Functional product MVP demo and development environment.
- Monitoring dashboard connected to product/services.
- Development Flow and Monitoring Flow diagrams matching implementation.
- Dataset source, preprocessing, benchmark method and measured evaluation results.
- Tools, technical decisions, trade-offs and verification evidence.
- Supabase schema screenshot and actual connection/persistence evidence.
- Screenshots of relevant system tabs, with secrets redacted.
- Source-code repository/handoff and pitch deck.
- Prepare all team members to explain the complete system.

Before submission: replace placeholders and Pending results with real project evidence. This report deliberately separates known design from unverified runtime claims.